

Process Characterization Master Report

A-Mab Drug Substance — Stage 1 Process Characterization — PCMR-001

Manufacturing Science & Technology (MSAT) / Process Development

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

BLA, Biologics License Application · CPP, critical process parameter · CQA, critical quality attribute · Cpk, process-capability index · DoE, design of experiments · DS, drug substance · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · KPP, key process parameter · LRV, log-reduction value · MVM, minute virus of mice · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PPQ, process performance qualification · QbD, Quality by Design · WC-CPP, well-controlled CPP · XMuvLV, xenotropic murine leukaemia virus.

Executive summary

This Process Characterization Master Report (**PCMR-001**) consolidates the Stage 1 (Process Design) characterization of the A-Mab drug-substance (DS) process, rolling up the 8 per-unit-operation reports (**PCR-003 ... PCR-010**) into an integrated statement of process understanding, parameter classification, process capability, viral clearance and the overall control strategy. The characterization was executed under the Process Characterization Master Plan (**PCMP-001**) and the per-unit-operation plans, on qualified scale-down models, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023), the Quality-by-Design principles of ICH Q8, Q9, Q10 and Q11 (International Council for Harmonisation 2009, 2023b, 2008, 2012) and the viral-safety expectations of ICH Q5A(R2) (International Council for Harmonisation 2023a), with the process model and quality-attribute framework of the A-Mab case study (CMC Biotech Working Group 2009).

Headline outcomes

- All 37 process parameters across the 8 unit operations were classified on the A-Mab continuum: **1 CPP, 20 WC-CPP, 10 KPP and 6 GPP** (21 critical parameters in total). Every classification is supported by data from the characterization studies.
- All 10 drug-substance CQAs meet their acceptance criteria at commercial scale with a process capability of **Cpk 1.51** (all CQAs Cpk 1.33), the minimum being the cumulative Viral clearance — MVM (parvovirus) clearance.
- The modular viral clearance meets its requirements with margin: cumulative **18.87 log (XMuLV)** and **10.03 log (MVM)**, delivered orthogonally across the low-pH inactivation, anion-exchange and virus-filtration steps.
- The integrated process delivers an overall DS yield of **83.2%** (~**54.6 kg** of drug substance at 75 g/L) from the 15,000-L production culture.
- Multivariate design spaces were established for the DoE unit operations; the production bioreactor forms the glycan, charge-variant and aggregate CQAs, and the purification train clears the process-related impurities and provides the viral clearance.

The A-Mab drug-substance process is demonstrated to be well understood, robust and in control across its 8 unit operations. The parameter classifications, design spaces, proven acceptable ranges and per-CQA control strategy support the Stage 2 operating ranges and the receiving-site validation (**PTP-001**); the process is **ready for Stage 2 (Process Performance Qualification)**.

1 Introduction

1.1 Purpose

This master report consolidates the per-unit-operation characterization reports into a single statement of the drug-substance process understanding and control strategy suitable to support a Biologics License Application. It summarizes, across the whole train, the process description and performance, the CQA outcomes, the parameter classification, the process capability, the viral clearance and the overall control strategy, and it establishes Stage-2 readiness. The detailed studies, effects, models and design spaces are in the per-unit-operation reports cited herein.

1.2 Regulatory and scientific basis

The characterization follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the enhanced (QbD) approach of ICH Q8, Q9, Q10 and Q11 (International Council for Harmonisation 2009, 2023b, 2008, 2012); viral clearance is credited per ICH Q5A(R2) (International Council for Harmonisation 2023a). Criticality is treated as a continuum (ICH Q9). The report consolidates the Stage 1 Process Design outputs of PDA TR 60 §3.11 across the unit operations.

2 Process description and performance

The A-Mab DS process comprises the 8 unit operations of Table 2.1, operated at 15,000 L with a nominal titre of 4.37 g/L. The process flow and the principal quality impact of each step are shown in Figure 2.1.

Table 2.1: The A-Mab drug-substance process train (Steps 3–10) and each step’s principal role in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

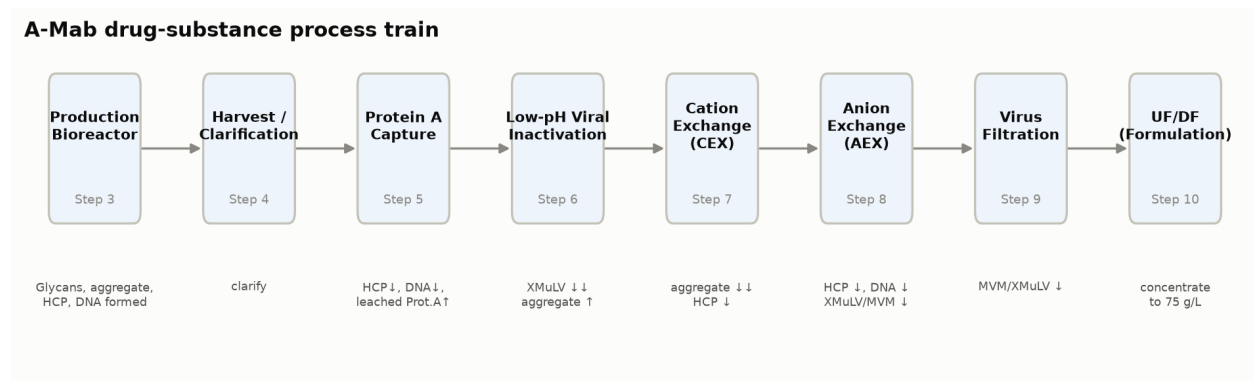


Figure 2.1: A-Mab drug-substance process flow (Steps 3–10) with the principal quality impact of each unit operation.

The integrated process delivers an overall DS yield of 83.2%, approximately 54.6 kg of drug substance at 75 g/L. The cumulative step-yield across the train is shown in Figure 2.2.

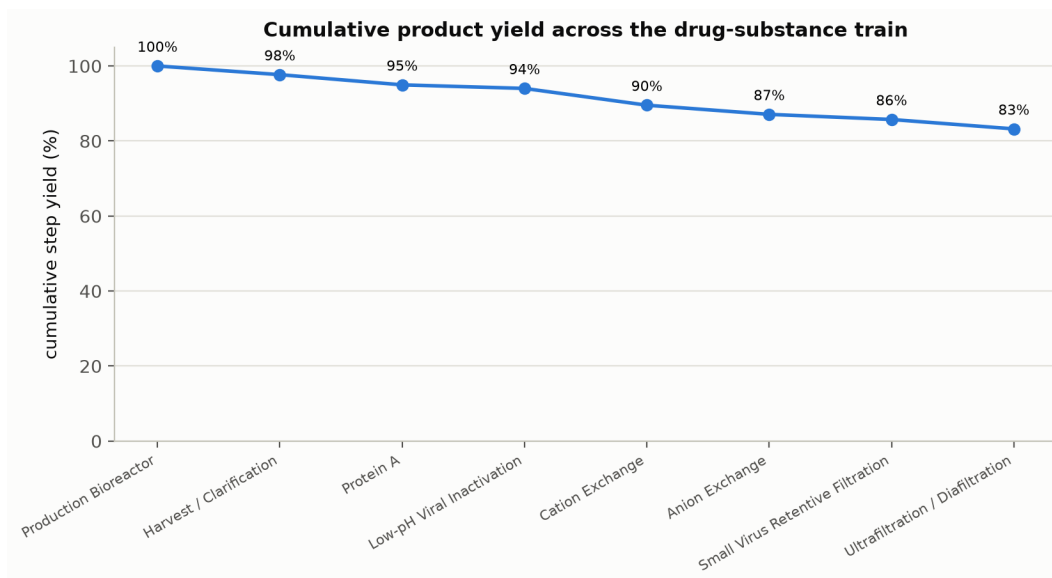


Figure 2.2: Cumulative product step-yield across the eight modelled unit operations.

3 Consolidated CQA outcomes

The 10 drug-substance CQAs, the unit operation that primarily sets or controls each, and the commercial-scale outcome (mean $\pm$ SD and capability against the acceptance criterion) are consolidated in Table 3.1. Every CQA meets its acceptance criterion with a capability of Cpk 1.51.

Table 3.1: Consolidated drug-substance CQA outcomes — the step that primarily sets/controls each CQA and the commercial-scale capability.

CQA	Crit.	Set/controlled by	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	Production Bioreactor	2–13	7.98 $\pm$ 0.58	2.89
Galactosylation (total %Gal)	H	Production Bioreactor	10–40	26 $\pm$ 1.6	2.86
High Mannose	VH	Production Bioreactor	3–10	5.99 $\pm$ 0.51	1.94
Aggregates (HMW)	H	Production Bioreactor	0–5	0.753 $\pm$ 0.088	16.11
Acidic charge variants (deamidation)	VL	Production Bioreactor	18–40	22.3 $\pm$ 1.4	4.26
Host Cell Protein (HCP)	M-H	Production Bioreactor	0–100	15.2 $\pm$ 4.6	6.14
Residual DNA	VL	Production Bioreactor	0–0.001	0.0001 $\pm$ 0	10.81
Leached Protein A	L-M	Protein A Chromatography	0–5	0.003 $\pm$ 0.0006	2785.30
Viral clearance — XMuLV (enveloped)	VH	Low-pH Viral Inactivation	16.7–30	19.5 $\pm$ 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	Anion Exchange Chromatography	8.6–20	10.4 $\pm$ 0.4	1.51

The glycan and charge-variant CQAs (afucosylation, galactosylation, high mannose, acidic variants) are formed and controlled in the production bioreactor within its multivariate design space; the aggregate is formed upstream, can rise during the low-pH hold, and is polished at cation exchange; the process-related impurities (HCP, residual DNA, leached Protein A) are formed upstream and cleared modularly across Protein A, cation exchange and anion exchange; and the viral clearance is delivered orthogonally across the low-pH inactivation, anion-exchange and virus-filtration steps.

4 Parameter classification summary

All 37 process parameters were classified on the A-Mab continuum (Table 4.1, Figure 4.1). Of these, 21 are critical (1 CPP and 20 WC-CPP), 10 are KPP and 6 are GPP. The single true CPP is the low-pH viral-inactivation pH — a narrow, high-consequence parameter — while the remaining critical parameters are well-controlled CPPs whose impact is reliably controlled within the operating region.

Table 4.1: Parameter-classification summary across the train (detail in the per-unit-operation reports).

Classification	Parameters
CPP	1
WC-CPP	20
KPP	10
GPP	6

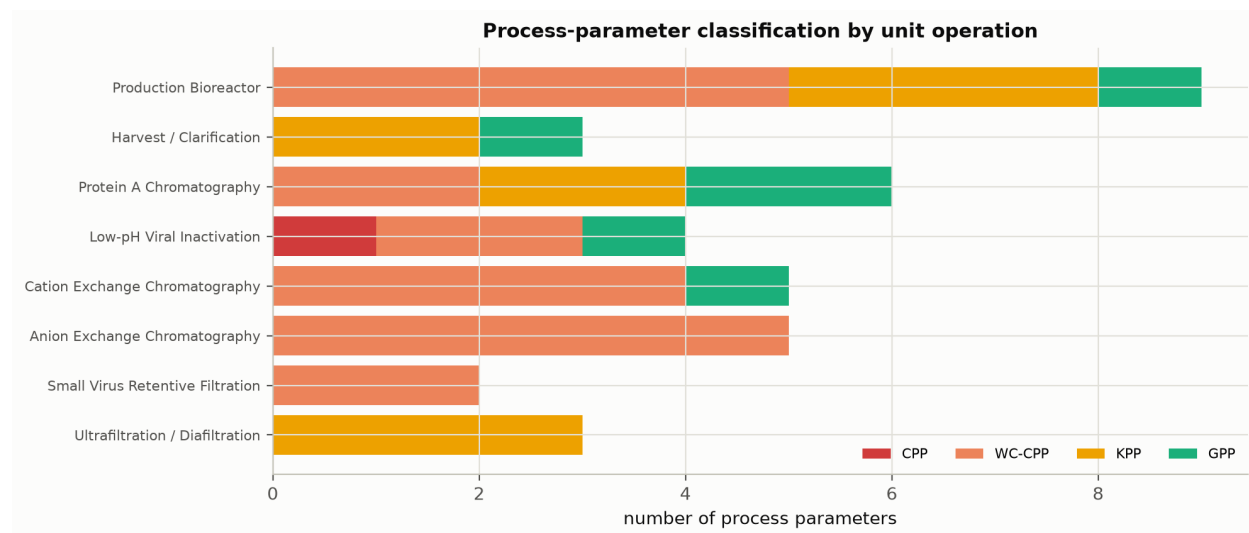


Figure 4.1: Post-characterization classification of process parameters (CPP / WC-CPP / KPP / GPP) by unit operation.

5 Process capability

Commercial-scale process capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter varied within its NOR (Figure 5.1). Every CQA meets its acceptance criterion, with the minimum capability at the cumulative Viral clearance — MVM (parvovirus) clearance (**Cpk = 1.51**); all CQAs achieve Cpk ≥ 1.33 . The capability is confirmed at commercial scale by the Stage-2 PPQ campaign (5 qualification batches as the capability basis).

Commercial-scale process capability (n = 2000 simulated batches)

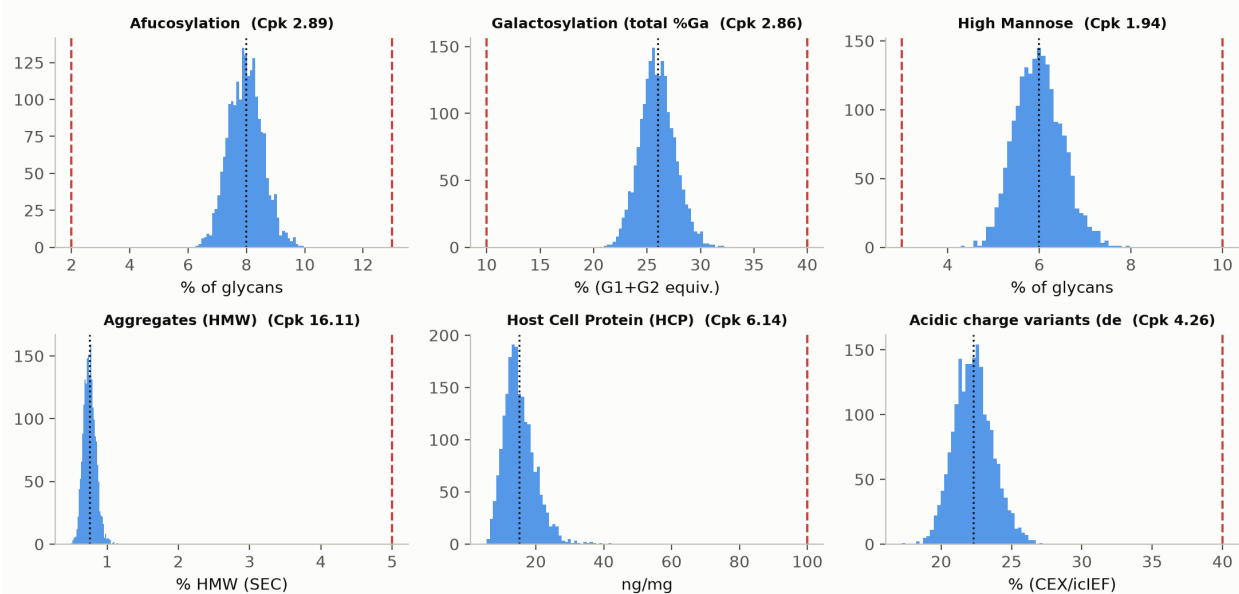


Figure 5.1: Monte-Carlo drug-substance CQA distributions vs acceptance limits, annotated with process capability (Cpk).

6 Viral clearance summary

The modular viral clearance is delivered by three orthogonal mechanisms — low-pH inactivation, anion-exchange adsorption and small-virus size-exclusion filtration — and credited per ICH Q5A(R2) (International Council for Harmonisation 2023a). The per-step and cumulative log-reductions are consolidated in Table 6.1 and Figure 6.1. The cumulative clearance is 18.87 log for XMuLV (enveloped retrovirus) and 10.03 log for MVM (small non-enveloped parvovirus), both above their cumulative requirements with margin.

Table 6.1: Modular viral clearance by step and cumulative (log), for XMuLV and MVM.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.10	0.00
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.87	10.03

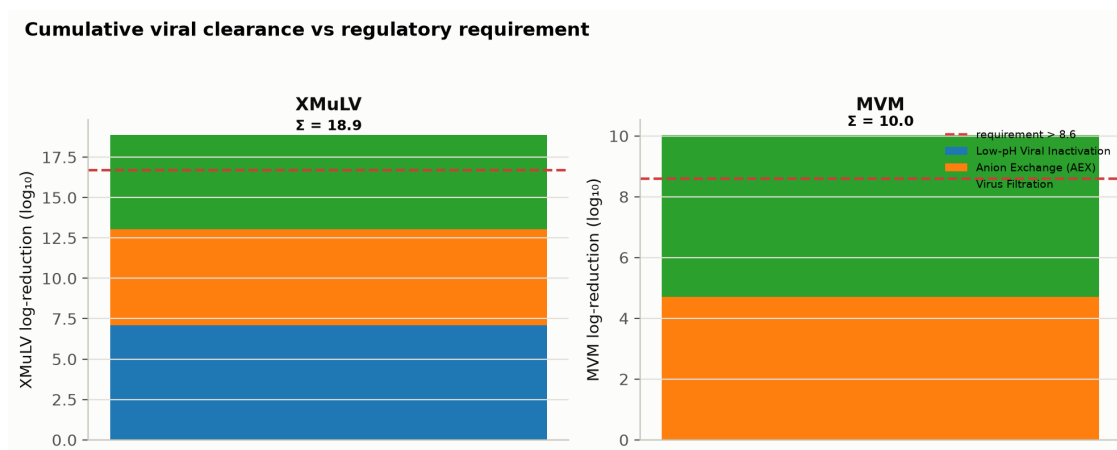


Figure 6.1: Modular viral clearance by step for XMuLV and MVM; cumulative clearance exceeds the total requirement with margin.

7 Overall control strategy

The overall control strategy is the sum of the individual per-CQA control strategies established across the unit operations. It comprises: control of the 1 CPP and 20 WC-CPPs to their NORs within the established design spaces, with enhanced control of the low-pH-inactivation pH; control of the 10 KPPs for process performance and of the 6 GPPs as general parameters; in-process controls and monitoring at each step; resin and membrane life-cycle control; the modular, orthogonal viral-clearance strategy; and the release and characterization testing of the 10 CQAs. The per-unit-operation contributions are detailed in the reports of Table 7.1 and are the basis of the drug-substance control strategy transferred to the commercial site (**PTP-001**).

Table 7.1: The per-unit-operation Process Characterization Reports consolidated in this master report.

Report	Step	Unit operation
PCR-003	3	Production Bioreactor
PCR-004	4	Harvest and Clarification
PCR-005	5	Protein A Chromatography
PCR-006	6	Low-pH Viral Inactivation
PCR-007	7	Cation Exchange Chromatography
PCR-008	8	Anion Exchange Chromatography
PCR-009	9	Small-Virus Retentive Filtration
PCR-010	10	Ultrafiltration / Diafiltration

8 Conclusions and Stage-2 readiness

The Stage 1 characterization demonstrates that the A-Mab drug-substance process is well understood, robust and in control across its 8 unit operations. All 37 parameters are classified with a data-supported rationale; multivariate design spaces are established for the DoE unit operations; all 10 CQAs meet their acceptance criteria at commercial scale with $Cpk \geq 1.51$; and the modular viral clearance meets its requirements with margin ($18.87 / 10.03 \log$ for XMuLV / MVM). The parameter classifications, design spaces, proven acceptable ranges and per-CQA control strategy support the Stage 2 operating ranges and the receiving-site validation. The process is **ready for Stage 2 (Process Performance Qualification)**; the post-characterization process risk assessment finalizes the CPP selection and the residual-risk position from these outcomes.

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